

# Robert Ortega

## Senior Data Engineer

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### SUMMARY

Senior Data Engineer with 8+ years of experience designing, developing, and deploying large-scale, cloud-based data solutions. Expert in **data modeling, ETL/ELT pipelines, real-time and batch data processing, and data warehousing** using Python, SQL, Spark, Snowflake, and Azure/AWS cloud platforms. Skilled in **streaming and messaging systems** such as Kafka and Kinesis, and experienced in **MLOps, DevOps, and automation** using Airflow, dbt, Docker, Kubernetes, and Terraform. Adept at building **scalable, high-performance data architectures**, ensuring **data quality, security, and operational efficiency**, and driving business insights across finance, healthcare, fintech, and eCommerce domains.

### EDUCATION

**Master's degree** in Data Science.

Johns Hopkins University      04/2014 – 10/2016

**Bachelor's degree** in Computer Science.

Illinois State University      08/2009 – 09/2013

### SKILLS

- **Programming & Scripting:** Python, PySpark, Scala, SQL, Bash, Java, TypeScript
- **Databases & Data Warehousing:** Snowflake, Redshift, Azure Synapse, Amazon RDS, PostgreSQL, MySQL, Oracle
- **Data Modeling:** 3NF, Star Schema, Snowflake Schema, Dimensional Modeling
- **Big Data & ETL/ELT Tools:** Hadoop, Hive, Sqoop, Spark Streaming, MapReduce, dbt, Meltano, Apache Airflow, AWS Glue, ELT pipelines
- **Streaming & Messaging:** Kafka, StreamSets, Kinesis
- **Cloud Platforms:** AWS (S3, EC2, Lambda, EMR, Glue, Redshift, Kinesis, CloudWatch, IAM, VPC), Azure (Data Lake, Databricks, HDInsight, Azure SQL, Data Factory, CosmosDB), GCP (BigQuery, Cloud Storage, AI VM)
- **DevOps & Orchestration:** Airflow, StreamSets, Meltano, Azure Data Factory, Docker, Kubernetes, Jenkins, Terraform, GitHub, Azure DevOps
- **Data Science & Machine Learning:** Pandas, NumPy, SciPy, Scikit-learn, MLlib, Spark ML, NLTK, predictive modeling, machine learning pipelines
- **Monitoring & Logging:** CloudWatch, Nagios, ELK Stack (Elasticsearch, Fluentd, Kibana)
- **Data Governance & Security:** Data profiling, data lineage, data validation, IAM policies, security best practices

#### **Soft Skills:**

- Strong problem-solving and analytical thinking
- Effective communication and cross-functional collaboration
- Creativity and innovation in AI/ML solutions
- Client-centric mindset and business impact focus
- Continuous learning and staying updated with AI/ML advancements

## EXPERIENCE

### Senior Data Engineer

I3 Verticals | Nashville, TN | Jan 2023 – Present

- Reviewed business requirements and composed source-to-target data mapping documents.
- Participated in **Agile/Scrum** development, actively contributing to sprint planning and meetings.
- Conducted **data profiling, merging, and transformation** from multiple structured and semi-structured data sources.
- Designed **3NF, Star, and Snowflake schemas** for OLTP, ODS, and dimensional data warehouses.
- Developed and optimized **data warehouses in Snowflake**, handling 100+ datasets and removing redundancy.
- Built **real-time and batch pipelines** using **Apache Kafka, Spark Streaming, and HDFS**, processing large-scale streaming data.
- Developed **ETL pipelines** using **Python, PySpark, Hive, Sqoop, MapReduce, Airflow** for workflow orchestration.
- Processed and stored **unstructured data** in HDFS, HBase, and MongoDB; scraped web data using **Beautiful Soup**.
- Applied **data validation, quality checks, and transformations** for reliable downstream analytics.
- Built and analyzed datasets with **Python (pandas, NumPy, SciPy, seaborn, matplotlib), Scikit-learn, NLTK**, performing feature engineering and ML modeling.
- Developed **machine learning pipelines** for predictive analytics using Random Forest, KNN, Linear/Multivariate Regression, Naive Bayes, and K-means clustering.
- Automated workflows using **Airflow, Cron scripts, Genie, and Databricks notebooks** for batch and real-time processing.
- Leveraged **Spark MLlib** to execute ML use cases on big data platforms.
- Worked with **Kafka clusters and StreamSets** for event log pipelines and real-time analytics.
- Monitored and visualized data pipelines using **Kibana, ELK Stack**, and cloud monitoring tools.
- Implemented **monitoring and logging** using **CloudWatch**, maintaining operational visibility and proactive issue resolution across ETL workflows.
- Leveraged **AWS Cloud services** to design and manage end-to-end data pipelines. Utilized **S3** for scalable storage of structured, semi-structured, and unstructured data, and **EC2** instances to run Spark clusters and Python workloads for large-scale processing.
- Orchestrated **big data processing on EMR** with Spark, Hadoop, Hive, and MapReduce for both batch and real-time workflows. Integrated **Kinesis** for real-time streaming ingestion and **Lambda** for serverless ETL and microservices automation.
- Used **Glue** for ETL orchestration and schema management, while **Redshift** and **RDS (PostgreSQL/MySQL)** handled analytical and transactional workloads.
- Ensured **pipeline reliability and monitoring** with **CloudWatch**, while managing secure access using **IAM roles and policies**.
- Designed and implemented **ETL/ELT pipelines** using **Azure Data Factory** to ingest, transform, and load data from multiple sources into **Azure SQL Database, Azure Synapse, and Data Lake**.
- Built **data pipelines** for structured and unstructured data, ensuring **data quality, consistency, and reliability** across production environments.
- Developed **data integration solutions** using **Azure Databricks** and **PySpark** for large-scale data processing and analytics.
- Deployed containerized workloads via **ECS and ECR**, and automated AWS infrastructure provisioning with **Terraform** for consistent, scalable, and cost-effective cloud deployments.

### Senior Data Engineer

Meta | Menlo Park, CA | Mar 2017 – Dec 2022

- Designed and built scalable **ETL/ELT pipelines** using **Python, PySpark, SQL**, and **Airflow** to process large-scale structured and unstructured data.

- Engineered **data warehouses and data marts** in **Snowflake, Redshift**, and SQL for analytics, reporting, and downstream consumption.
- Developed **real-time and batch data processing pipelines** using Kafka, Kinesis, Flume, and Spark Streaming for high-volume event streams.
- Managed and automated **data infrastructure in AWS** (S3, EC2, Lambda, Glue, Redshift, EMR, CloudFormation) and GCP (BigQuery, AI/ML VMs) for reliable data ingestion and processing.
- Implemented **data collection, ingestion, and transformation strategies** from APIs, databases, and cloud storage into analytical platforms.
- Built microservices and **Python-based data processing frameworks** (FastAPI, Flask) to support modular and reusable data pipelines.
- Designed **data modeling solutions** using 3NF, Star, and Snowflake schemas for OLTP, ODS, and OLAP systems.
- Automated **data quality checks and validations**, ensuring accurate and reliable reporting across multiple data sources.
- Optimized **data pipelines and storage** for performance, reducing ETL processing time and improving system throughput by 25%.
- Configured **CI/CD pipelines** with Jenkins, Terraform, Docker, and Kubernetes for automated deployment of data pipelines and processing workflows.
- Monitored **data pipelines and infrastructure** using CloudWatch, Nagios, and ELK Stack for proactive troubleshooting and performance tracking.
- Collaborated with data analysts and software engineers to design **scalable data solutions**, enabling analytics, ML, and business intelligence.
- Implemented **data visualization solutions** using Tableau and Power BI to support self-service analytics and actionable insights.

## Software Engineer (Data & Web)

Infosys | Lisle, IL | Oct 2016 – Feb 2017

- Developed **intelligent HR and CRM platforms** using **Python (Django, Flask, FastAPI)** for automating customer management, bookings, and reporting.
- Optimized complex **SQL queries** to fetch over **100M+ rows in milliseconds**, boosting database efficiency.
- Designed and deployed **microservices** and **RESTful APIs** using **Docker** and **Kubernetes** on **AWS (EC2, S3, Lambda, RDS, CloudWatch)**.
- Developed **microservices architecture** integrating **Python APIs** with **TypeScript Node.js services**, enabling modular, scalable model deployments.
- Automated CI/CD pipelines using **AWS CodePipeline**, **Jenkins**, and **GitHub Actions** to streamline releases.
- Built scalable **data analytics dashboards** with **React.js** and backend integration through **Flask APIs**.
- Leveraged **AWS Glue**, **Athena**, and **Redshift** for large-scale data ingestion, transformation, and querying.
- Implemented **asynchronous data processing** with **Celery** and **Redis**, improving response time by 30%.
- Managed serverless deployments using **AWS Lambda** and **API Gateway** for low-cost, scalable architecture.
- Developed **unit tests** using **PyTest** and automated monitoring with **AWS CloudWatch** and **ELK Stack**.
- Built scalable **data analytics dashboards** with **React.js** integrated with **Flask APIs** for backend data access.
- Implemented **asynchronous data processing** with **Celery** and **Redis**, improving response times by 30%.
- Implemented **real-time streaming data pipelines** using **Kafka**, **Spark Streaming**, **Kinesis**, and **Lambda** functions for event-driven workflows.
- Collaborated in **Agile/Scrum** sprints to deliver production-grade, cloud-native Python applications.

## CERTIFICATIONS & REWARDS

- AWS Data Engineering Nanodegree Udacity
- Data Analyst Nanodegree Udacity

- Monash Merit Scholarship
- Machine Learning Bootcamp Udemy
- Python for Data Science Udemy